

Visualization Literacy in United States Classrooms: A Qualitative Analysis of State K–12 Learning Standards

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Abstract

Visualization literacy, or the ability to understand and interpret graphical displays of data, is an essential skill for everyday decision-making. While prior research has focused on assessing visualization literacy in adults, less is known about how these skills are developed through K–12 education. In the United States, state-level learning standards define what students are expected to learn, providing a structured view of how visualization literacy is taught in schools. In this work, we collect STEM education standards from all 50 U.S. states and extract over 5,000 visualization-related standards from a representative sample of 20 states. Through thematic analysis, this work will enable the creation of a framework describing how visualization literacy is taught across grade levels and mapped to existing visualization literacy frameworks. This work will connect visualization literacy theory with formal education standards and identify gaps between current theory and classroom expectations.

Research Questions

1. Do K–12 education standards include visualization literacy skills that are not represented in existing task-based visualization literacy frameworks?
2. What visualization tasks are students expected to perform at each grade, and what idioms are they expected to be familiar with?

Methodology

- Collected K–12 STEM standards from all 50 states; selected 20 representative states for analysis.
- Developed inter-rater reliability (IRR) by independently identifying visualization literacy standards from documents.
- Extracted and grouped visualization literacy standards using text cleaning and fuzzy string matching.
- Independently coded standards using the visualization literacy codebook and grouping, then compared and reconciled codes.
- Produced a dataset of 5,000+ standards and an emerging visualization literacy framework.

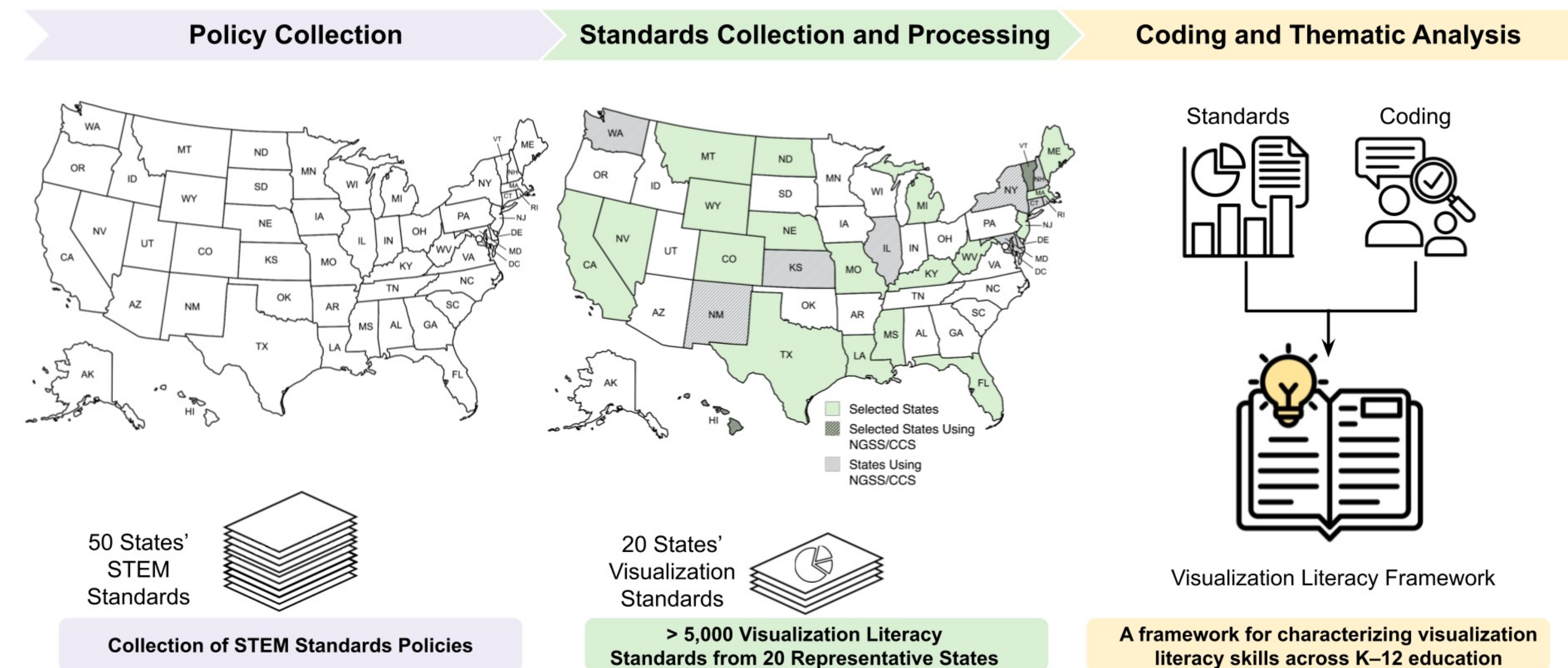
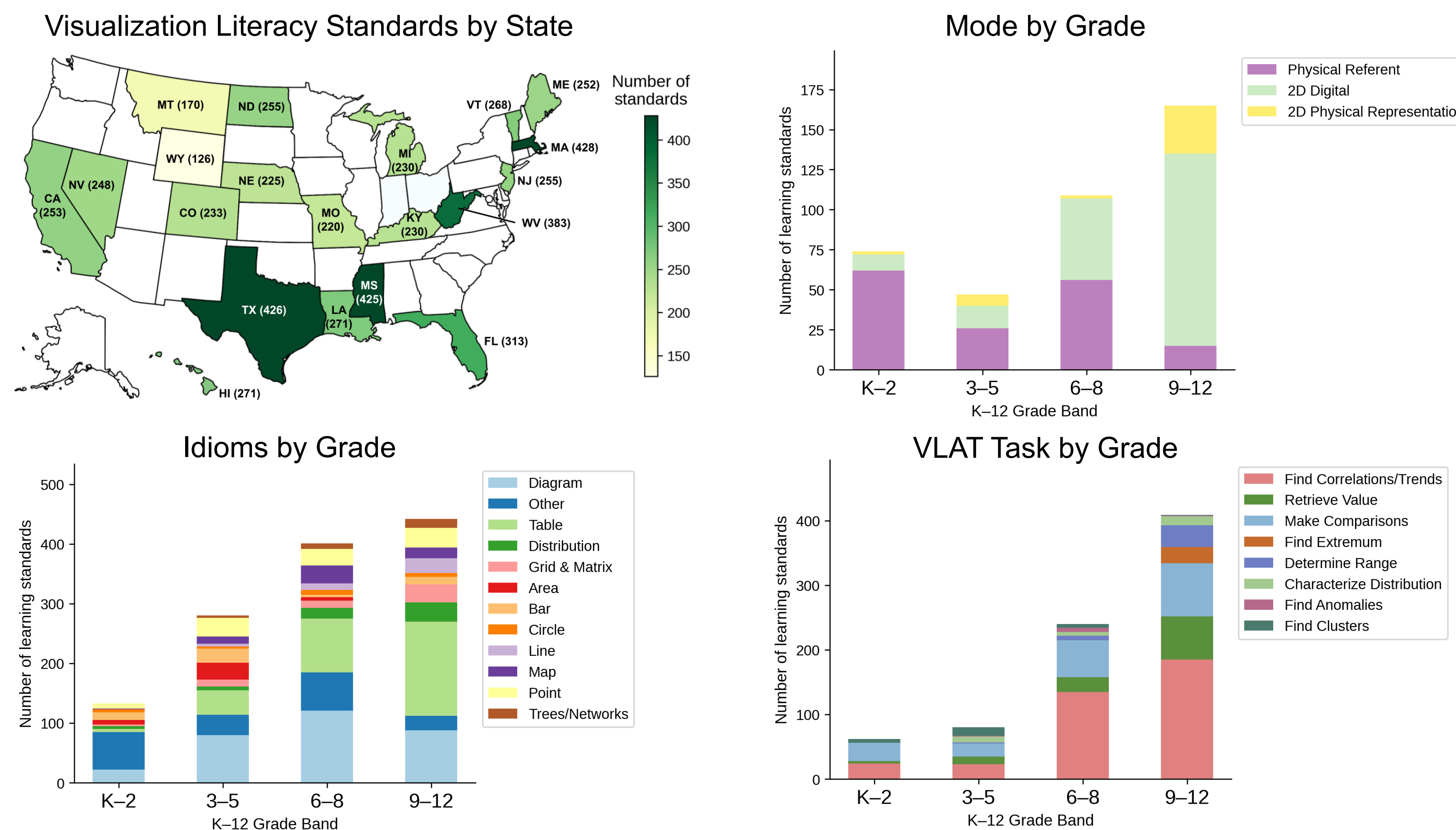


Figure 1: Overview of the methodology used to construct the K–12 Visualization Literacy Standards dataset. STEM education standards were collected from all 50 U.S. states, and a representative subset of 20 states was selected for detailed analysis. Visualization-related standards were identified and analyzed using similarity clustering and qualitative coding¹ to develop a visualization literacy framework.



Figures 2–5: Choropleth map of total visualization literacy standards by state (top left); stacked bar charts showing the distribution of visualization idioms² (bottom left), representation modes (top right), and VLAT tasks³ (bottom right) across K–12 grade bands.

“Construct and interpret scatter plots for bivariate numerical data to investigate patterns of association between two quantities. Describe patterns such as clustering, outliers, positive or negative association, linear association and nonlinear association.” Kentucky Math, Grade 8 (8.SP.1)

“Diagram, utilizing digital or physical models, the water cycle and how it relates to the total amount of fresh water available to living things at any given time.” Mississippi Science, High School (MAQ.1.3)

Figure 6: Example standards from the dataset, with visualization idioms and tasks highlighted in yellow.

Results

- Visualization literacy standards are most prevalent in grades 9–12.
- Retrieval, comparison, and trend-finding are the most common tasks across all grade levels
- Physical referents dominate early grades, while 2D digital representations increase in middle and high school.
- Tables and diagrams are the most common visualization idioms across subjects.
- Most standards do not specify the idiom used.

Next Steps and Future Works

- Develop and validate a framework mapping visualization literacy skills to grade bands.
- Interview educators to understand how standards translate to classroom practice.
- Expand scope to include non-STEM subjects, College Board’s AP standards, and district-level case studies.

Acknowledgments and References

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- [1] A. Bryman and B. Burgess, Eds., *Analyzing Qualitative Data*, 1st ed. London, UK: Routledge, 1994, doi: 10.4324/9780203413081.
- [2] M. A. Borkin, A. A. Vo, Z. Bylinskii, P. Isola, S. Sunkavalli, A. Oliva, and H. Pfister, “What makes a visualization memorable?,” *IEEE Trans. Vis. Comput. Graphics*, vol. 19, no. 12, pp. 2306–2315, Dec. 2013, doi: 10.1109/TVCG.2013.234.
- [3] S. Lee, S.-H. Kim, and B. C. Kwon, “VLAT: Development of a Visualization Literacy Assessment Test,” *IEEE Trans. Vis. Comput. Graphics*, vol. 23, no. 1, pp. 551–560, Jan. 2017, doi: 10.1109/TVCG.2016.2598920.